Trigonometry: Double Angle Formulae

Questions 1–38

Question 1. Use the Compound Angle Formulae $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ to find the Double Angle Formula for $\sin 2x$.

Question 2. Use the Compound Angle Formulae $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$ to find a simplified form for $\cos 2x$.

Question 3. Write down $\cos 2x$ just in terms of $\cos x$.

Question 4. Write down $\cos 2x$ just in terms of $\sin x$.

Question 5. Use the Compound Angle Formulae $\tan(A \pm B) = \dfrac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$ to find a simplified form for $\tan 2x$.

Question 6. Use $\tan 2x = \frac{\sin 2x}{\cos 2x}$ to find the same result found in question 5.

Question 7. Simplify the following expressions:

a) $2\sin 2x\cos 2x$

b) $2\cos^2 3x - 1$

c) $1 - 2\sin^2 4x$

d) $4\sin\frac{1}{2}x\cos\frac{1}{2}x$

e) $6\cos^2\frac{1}{3}x - 6\sin^2\frac{1}{3}x$

f) $6\cos^2\frac{3}{2}x - 3$

Question 8. Solve the following equations:

a) $\sin 2\theta - 2\sin\theta = 0$ for $0 \leq \theta < 360°$

b) $\sin 2\theta + 5\sin\theta = 0$ for $0 \leq \theta < 2\pi$

c) $3\sin 2\theta = 2\cos\theta$ for $0 \leq \theta < 360°$

d) $8\sin 2\theta + 5\cos\theta = 0$ for $0 \leq \theta < 2\pi$

Question 9. Solve the following equations:

a) $\cos 2\theta + \cos\theta = 0$ for $0 \leq \theta < 360°$

b) $3\cos 2\theta = 7\cos\theta$ for $0 \leq \theta < 2\pi$

c) $2\cos 2\theta + 2\sin\theta = 0$ for $0 \leq \theta < 360°$

d) $2\cos 2\theta = 9\sin\theta$ for $0 \leq \theta < 2\pi$

Question 10. Prerequisite knowledge for Q10: Reciprocal Trigonometric Functions Solve the following equations:

a) $2\sin 2\theta \operatorname{cosec}\theta = 1$ for $0 < \theta < 360°$

b) $5\sin 2\theta \sec\theta = 2$ for $0 < \theta < 2\pi$

c) $\cos 2\theta \operatorname{cosec}\theta = 3$ for $0 < \theta < 360°$

d) $9\cos 2\theta \sec\theta = 4$ for $0 < \theta < 2\pi$

Question 11. It is given that $\sin x = \frac{2}{5}$ and that x is acute.
Find the exact value of $\sin 2x$.

Question 12. It is given that $\sin x = \frac{3}{8}$ and that x is acute.
Find the exact value of $\cos 2x$.

Question 13. It is given that $\cos x = -\frac{2}{9}$ and that x is obtuse.
Find the exact value of $\tan 2x$.

Question 14. Given that $10\sin^2 x + 5\sin 2x = 4$, show that $3\tan^2 x + 5\tan x - 2 = 0$.

Question 15. Solve the following equations:

a) $34\sin^2\theta - \sin 2\theta = 30$ for $0° \leq \theta < 360°$

b) $22\cos^2\theta - 13\sin 2\theta = 10$ for $0 < \theta < 2\pi$

Question 16. Prerequisite knowledge for Q16–Q21: Reciprocal Trigonometric Functions Prove the following trigonometric identities:

a) $\tan 2x + \cot x \equiv \cot x \sec 2x$

b) $\tan 2x + \sin 2x \equiv \tan 2x(\cos 2x + 1)$

c) $\tan 2x + \cot 2x \equiv 2\operatorname{cosec} 4x$

Question 17. Prove the following trigonometric identities:

a) $\cot 2x - \cos 2x \equiv \cot 2x(1 - \sin 2x)$

b) $\cot 2x + \operatorname{cosec} 2x \equiv \operatorname{cosec} 2x(1 + \cos 2x)$

c) $\cot 2x + \operatorname{cosec} 2x \equiv \cot x$

Question 18. Prove the following trigonometric identity:

$$\tan 2x + \operatorname{cosec} 2x \equiv 2 \operatorname{cosec} 4x(\sin^2 2x + \cos 2x)$$

Question 19. Prove the following trigonometric identities:

a) $\frac{2\tan 2x}{1+\tan^2 2x} \equiv \sin 4x$

b) $(\sin 2x + \cos 2x)^2 \equiv 1 + \sin 4x$

c) $\frac{2\sin x}{\cos 2x+1} - \frac{2\sin x}{\cos 2x-1} \equiv \operatorname{cosec} x \sec^2 x$

Question 20. Prove the following trigonometric identities:

a) $\frac{1-\sin 2x}{1+\sin 2x} \equiv \frac{2}{(\sin x+\cos x)^2} - 1$

b) $\frac{1-\cos 2x}{1+\sin 2x} \equiv \frac{2\sin^2 x}{(\sin x+\cos x)^2}$

c) $\frac{1+\cos 2x-\sin 2x}{1+\sin 2x-\cos 2x} \equiv \frac{\cos x(\cot x-1)}{\sin x+\cos x}$

Question 21.

a) Prove the following trigonometric identity:

$$\operatorname{cosec}^2 x \equiv \frac{\cos 2x + 3\sin^2 x + \cos^2 x}{2\sin^2 x}.$$

b) Hence solve

$$(\cot^2 \theta - 1)\operatorname{cosec}^2 \theta = \frac{\cos 2\theta + 3\sin^2 \theta + \cos^2 \theta}{6\sin^2 \theta}$$

for $0 < \theta < 2\pi$.

Question 22.

a) Solve

$$11\sin 2x = 2\cos x + 6\cos^3 x$$

for $0 < x < 2\pi$.

b) Solve

$$11\sin(6\theta - \tfrac{\pi}{2}) = 2\cos(3\theta - \tfrac{\pi}{4}) + 6\cos^3(3\theta - \tfrac{\pi}{4})$$

for $0 < \theta < \frac{2\pi}{3}$.

Question 23. Prove the following trigonometric identities:

a) $\sqrt{2}\sin(2x + \frac{\pi}{4}) \equiv \cos^2 x + 2\sin x \cos x - \sin^2 x$

b) $2\sqrt{3}\sin(\frac{\pi}{3} - 2x) \equiv 3\cos^2 x - 2\sqrt{3}\sin x \cos x - 3\sin^2 x$

c) $\sin(\frac{\pi}{4} - 2x) + \cos(2x + \frac{\pi}{4}) + 2\sqrt{2}\sin^2 x \equiv \sqrt{2}(\cos x - \sin x)^2$

d) $\frac{1}{2}\cos(\frac{\pi}{6} - 2x) - \frac{1}{2}\sin(\frac{\pi}{3} - 2x) \equiv \sin x \cos x$

Question 24. Prove the following trigonometric identity:

$$(\sin(x + \tfrac{\pi}{4}) + \cos(x + \tfrac{\pi}{4}))^2 \equiv \cos 2x + 1$$

Question 25. Prove the following trigonometric identity:

$$(\sin(2x + \tfrac{\pi}{4}) - \cos(2x + \tfrac{\pi}{4}))^2 \equiv 1 - \cos 4x$$

Question 26. Use the Compound Angle Formulae $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ to show that $\sin 3x \equiv 3 \sin x - 4 \sin^3 x$.

Question 27. Use the Compound Angle Formulae $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$ to show that $\cos 3x \equiv 4\cos^3 x - 3\cos x$.

Question 28. Prove the following trigonometric identity:

$$\sqrt{2}\sin(\tfrac{\pi}{4} - x)(2\sin 2x + 1) \equiv \sin 3x + \cos 3x$$

Question 29. Find the period of $\mathrm{f}(x) = 5 - \cos^2 x$, $x \in \mathbb{R}$.

Question 30. Prove the following trigonometric identity:

$$\frac{\sin 3x}{\sin x} + \frac{\cos 3x}{\cos x} \equiv 4\cos 2x$$

Question 31. Use the Compound Angle Formulae $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$ to find $\tan 3x$ in terms of $\tan x$ only.

Question 32. Solve the equation $\tan 3\theta + \tan \theta = 0$ for $0° < \theta < 360°$.

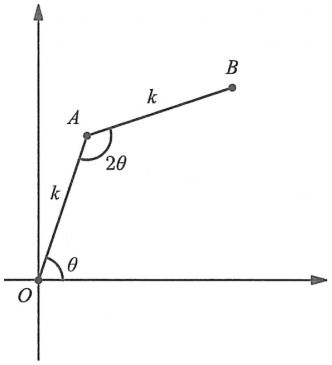
Question 33. Solve $\sin\frac{2\theta}{3}\cos\frac{2\theta}{3} = \frac{1}{2}$ for $0 < \theta < 2\pi$.

Question 34. Solve $\cos^2\frac{3\theta}{2} - \sin^2\frac{3\theta}{2} = \frac{1}{3}$ for $0° < \theta < 360°$.

Question 35. The diagram below shows the rod OAB which is pivoted at O. The lengths of OA and AB are both k units. Taking O as the point with coordinates $(0, 0)$,

a) show that the coordinates of B are $(4k\sin^2\theta\cos\theta,\ 2k\sin\theta - 4k\sin\theta\cos^2\theta)$

b) Find the distance of B from O when $\theta = 60°$ in terms of k.



Question 36. Prove the following trigonometric identity:

$$\sin 4x \equiv 4\sin x\cos^3 x - 4\sin^3 x\cos x$$

Question 37. Prove the following trigonometric identity:

$$\cos 4x \equiv 8\cos^4 x - 8\cos^2 x + 1$$

Question 38. Prerequisite knowledge for Q38: Reciprocal Trigonometric Functions Prove the following trigonometric identity:

$$\frac{\sin 4x}{\cos x} + \frac{\cos 4x}{\sin x} \equiv (2\cos 2x - 1)\operatorname{cosec} x$$